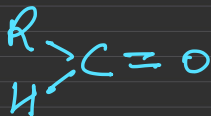


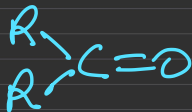
Aldehyde, Ketone & Carboxylic Acid

Organic compounds having carbon-oxygen double bond are known as carbonyl compounds. Carbonyl compounds are of two types 1.) Aldehyde 2.) Ketone

Aldehyde

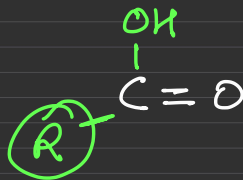


Ketone



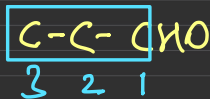
Carboxylic Acid

Carbonyl Hydroxyl



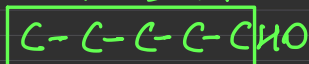
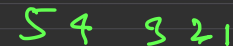
Aldehyde ($-CHO$)

$-al$ [if "C" of $-CHO$ is included in chain]

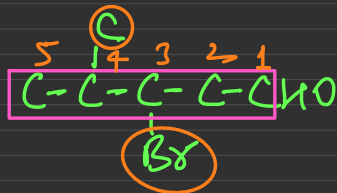


Propan-1-al

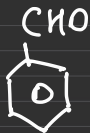
$-carbaldehyde$ (if "C" of $-CHO$ is not included in chain)



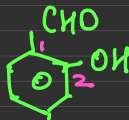
Pentan-1-al



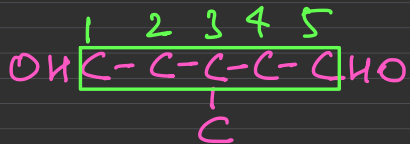
3-Bromo-4-methyl pentan-1-al



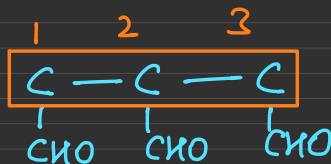
Benzene carbaldehyde
or Benzaldehyde



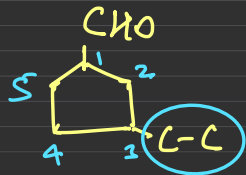
2-Hydroxy
benzaldehyde
(Salicylaldehyde)



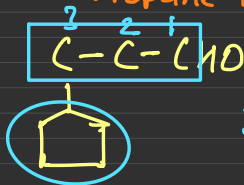
3-Methylpentane-1,5-dial



Propane-1,2,3-tricarbaldehyde



3-Ethylcyclopentane-1-
carbaldehyde

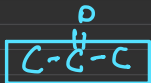


3-Cyclopentyl
propan-1-al

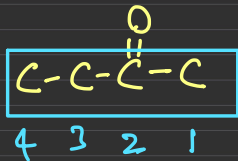
Ketone



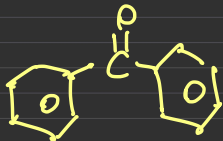
one



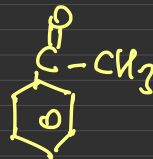
Propan-2-one



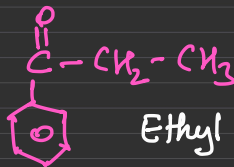
Butan-2-one



Diphenyl ketone
or Benzo phenone



Methyl phenyl ketone
or Acetophenone



Ethyl phenyl ketone
or propio phenone

METHODS OF PREPARATION:

By Oxidation of Alcohol

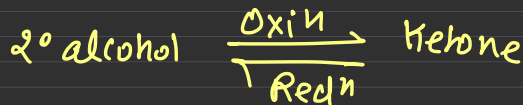
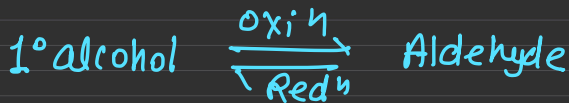
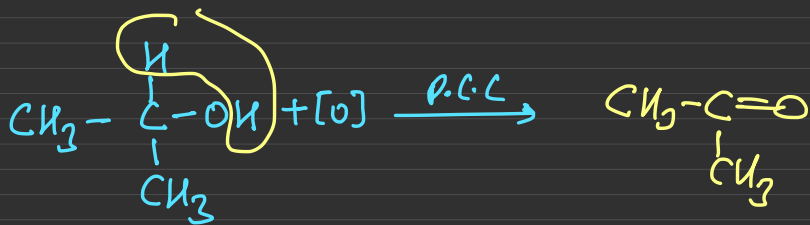
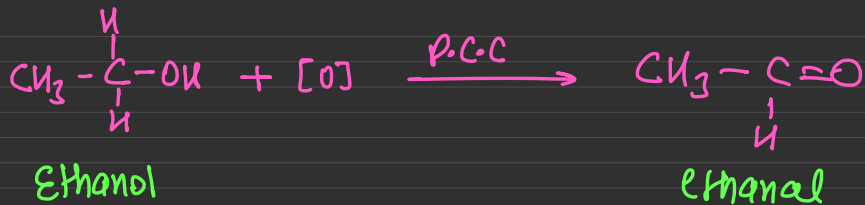
P.C.C $\rightarrow$ Pyridinium Chloro Chromate

1° alcohol $\xrightarrow{\text{P.C.C}}$ Aldehyde

2° alcohol $\xrightarrow[\text{CH}_2\text{Cl}_2]{\text{P.C.C in } \text{P}}$ Ketone

$\downarrow$

Corey's Reagent



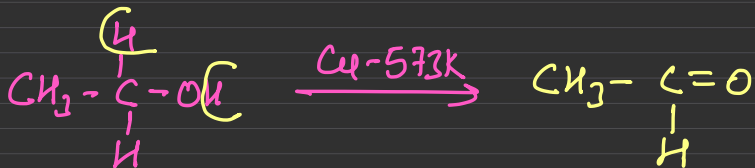
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DEHYDROGENATION

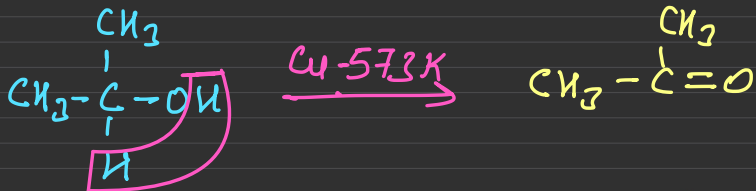
→

$\text{Cu} - 573\text{K}$

1° alcohol $\xrightarrow[\text{Cu} - 573\text{K}]{\text{Dehydrogenation}}$ Aldehyde



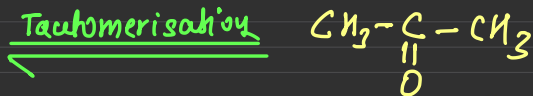
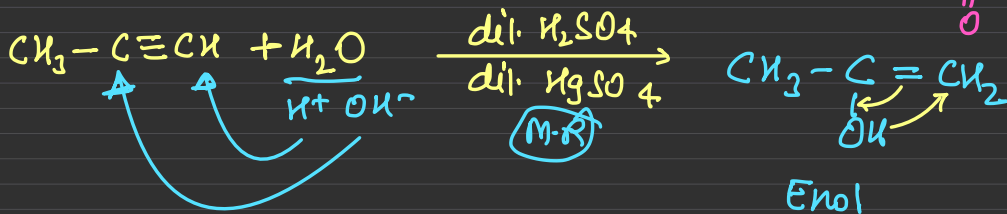
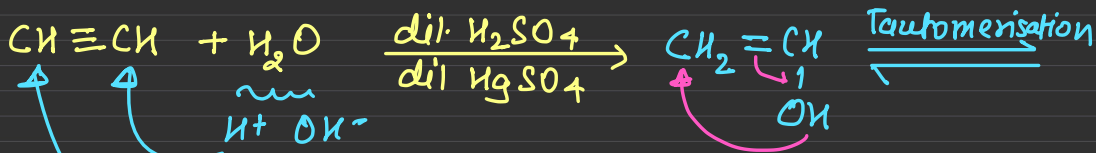
2° alcohol $\xrightarrow[\text{Cu} - 573\text{K}]{\text{Dehydrogenation}}$ Ketone



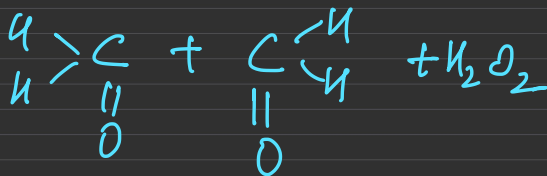
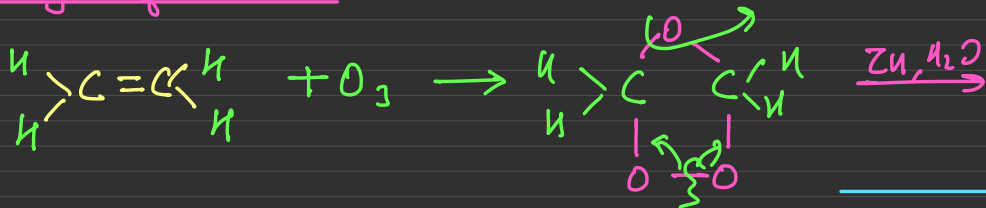
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FROM HYDROCARBONS

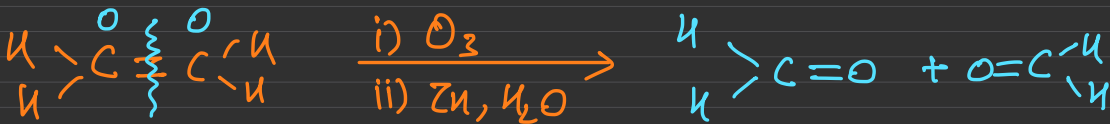
↳ From Alkynes

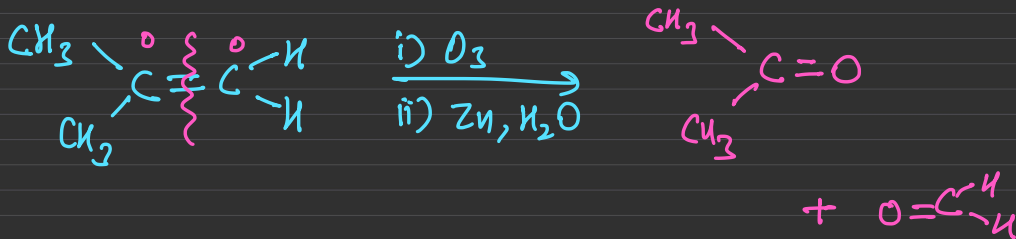
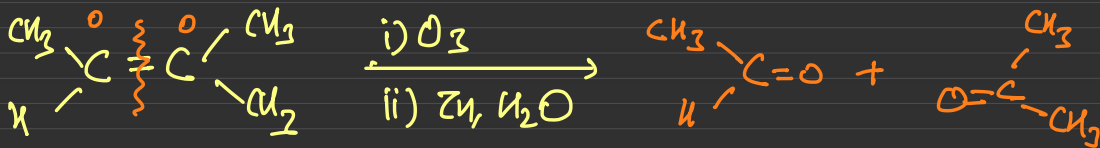


Ozonolysis of Alkenes



double Bond
cut cuts cut
along with [O]
cut cuts cut

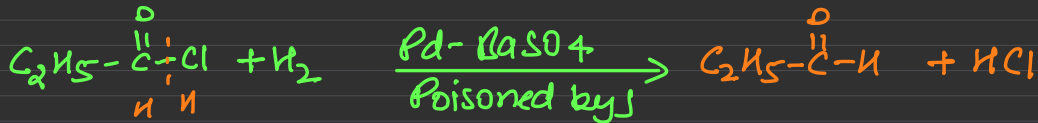
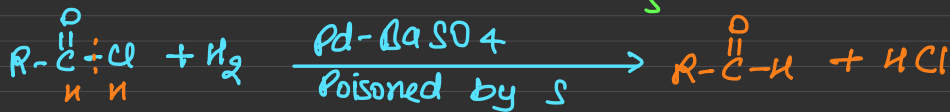




Preparation of Aldehyde Only

• Rosenmund Reduction

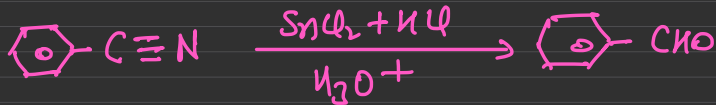
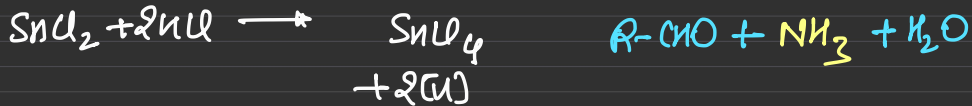
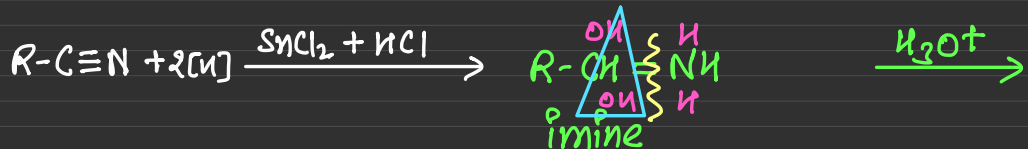
Acid chloride + H_2 $\xrightarrow[\text{Poisoned by 'S'}]{\text{Pd-BaSO}_4}$ Aldehyde



Note The poisoning of catalyst with sulphur to decrease its activity. So the reduction stops at aldehyde but if poisoning is not done then acid chloride will be reduced to alcohol.

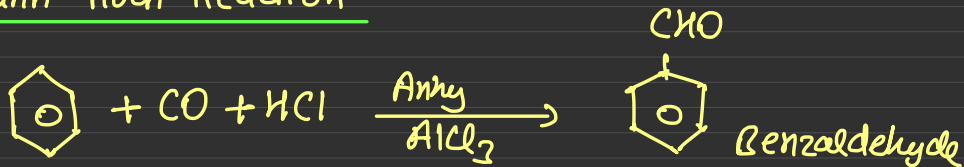
STEPHEN REDUCTION:

Stephen Cyana Aldehyde बना गया
सोने लकले हल्का पानी डबाल

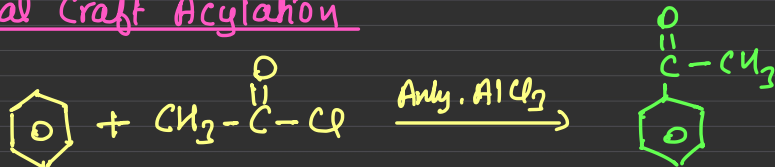


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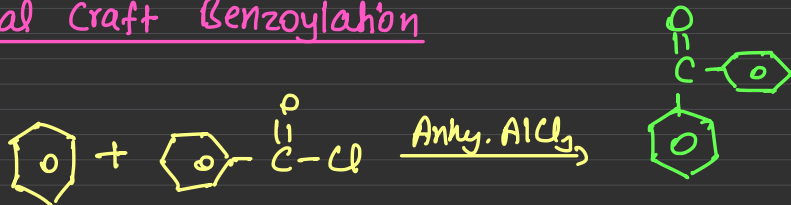
Gattermann Koch Reaction



• Friedel Craft Acylation



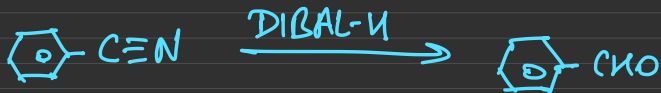
• Friedel Craft Benzoylation



DIBAL-H

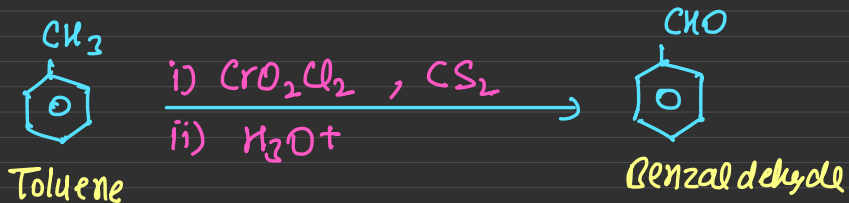
Di isobutyl aluminium hydride

→ nitrile
→ ester  Aldehyde



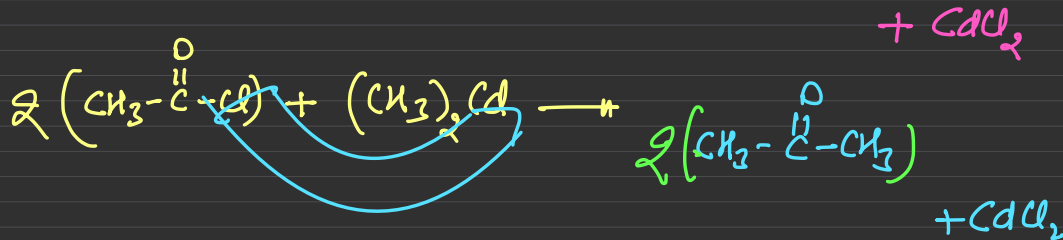
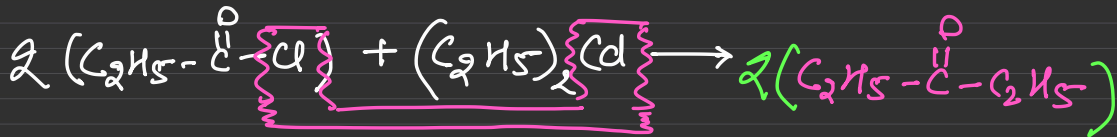
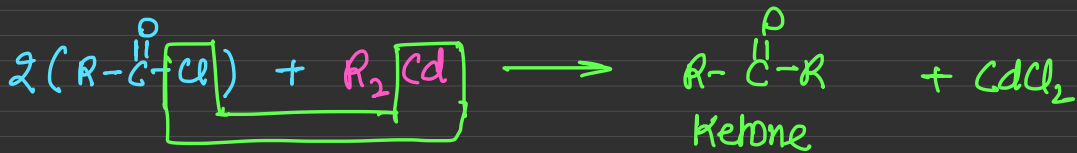
ETARD REACTION (99%)

Trick - तौतले Benzal crow मिलेगा चल

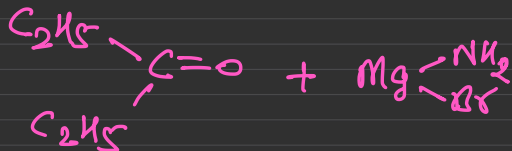
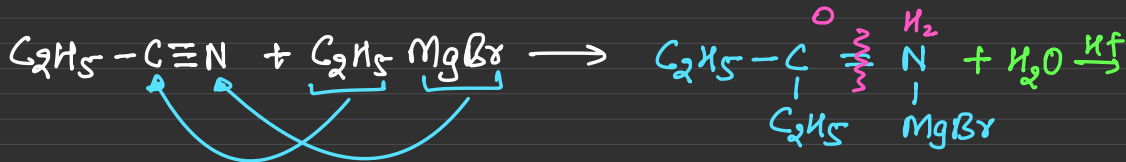
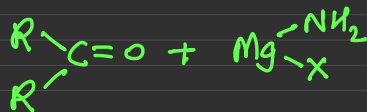


$\text{CrO}_2\text{Cl}_2 \rightarrow$ Chromyl Chloride

Preparation of Ketone Only :-



FROM GRAIGNARD REAGENT :



PHYSICAL PROPERTIES

↳ PHYSICAL STATE

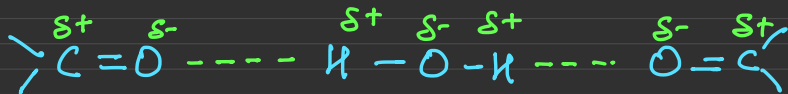
Methanal is a gas

Ethanal is a low boiling volatile liquid

while other aldehydes upto 11 carbons are colourless liquid.

All ketones are colourless liquid

↳ Solubility

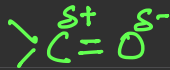


aldehyde and ketones upto 4 carbons are soluble in water due to the formation of intermolecular H-Bond.

Higher aldehyde and ketones are insoluble in water due to large hydrophobic part.

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Boiling Point

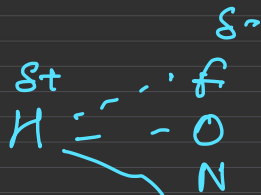
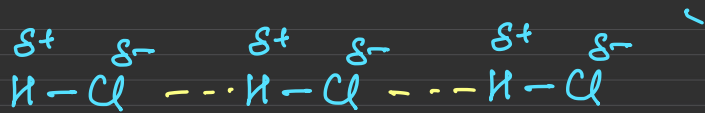


B.Pt of aldehyde and ketones are higher than hydrocarbons of comparable molecular mass, because carbonyl groups are polar in nature so have dipole dipole interactions.

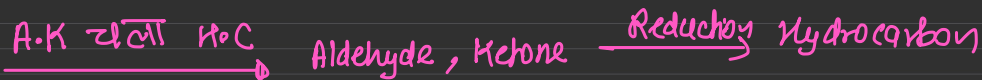
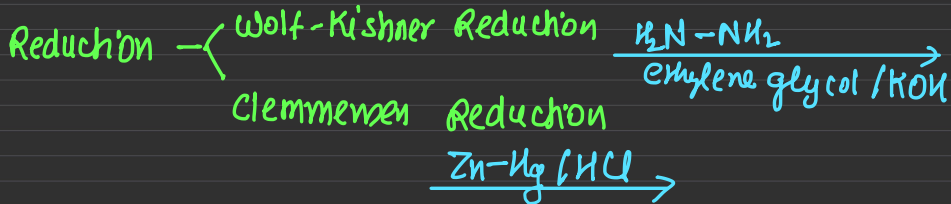
They have less b.pt than alcohol due to absence of intermolecular hydrogen bonding.

among aldehyde and ketone, ketone have higher b.pt because ketones are more polar hence have higher dipole moment

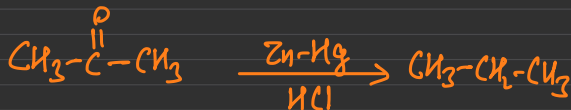
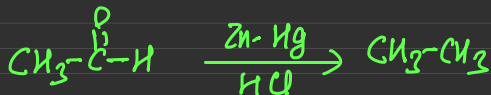




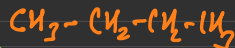
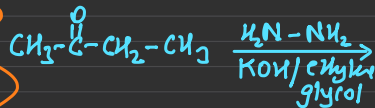
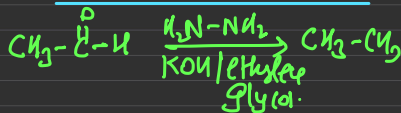
Chemical Properties of Aldehydes & Ketones



Clemmensen Redn

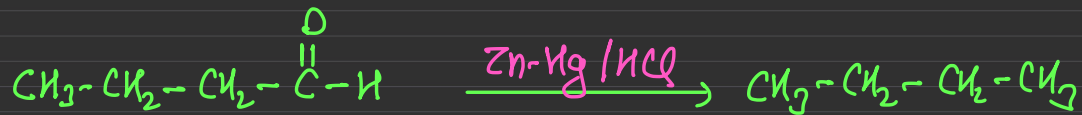


Wolf Kishner Redn



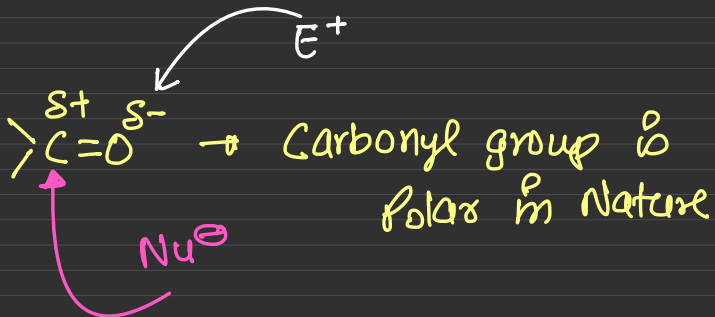
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Butanal $\longrightarrow$ Butane

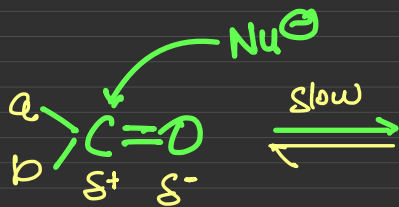


99%

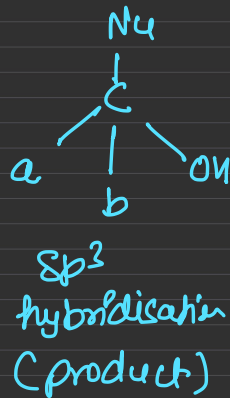
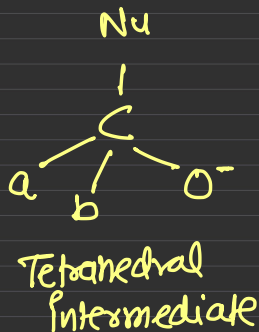
NUCLEOPHILIC ADDITION REACTION



Mechanism of Nucleophilic Addition Rxn



sp^2
hybridisation



99%

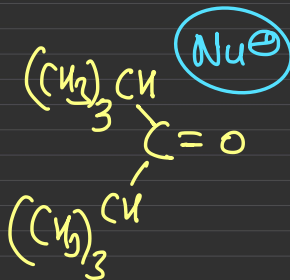
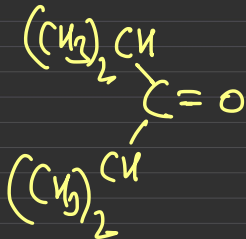
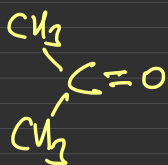
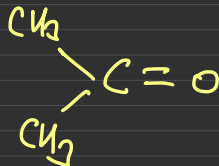
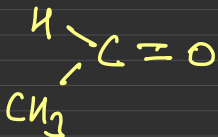
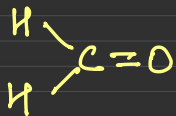
Reactivity order of aldehyde and ketones towards Nucleophilic Addition Reaction

< **Steric Reason** Among Aldehyde and ketones, aldehydes
< **Electronic Reason** are more reactive than ketone because
in aldehyde only one alkyl group is present hence
have less +I effect

On the other hand in ketones there are
two alkyl group. So more +I effect. decrease the
electrophilicity of carbonyl carbon and hence making it
less available for attacking by nucleophile

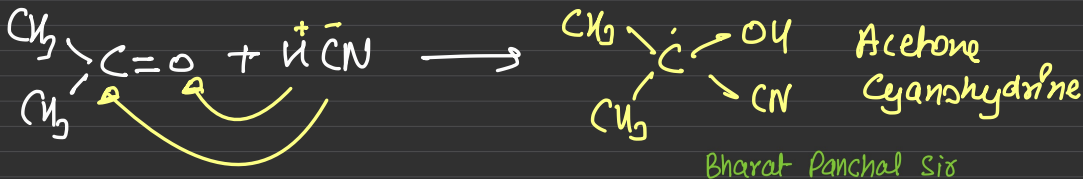
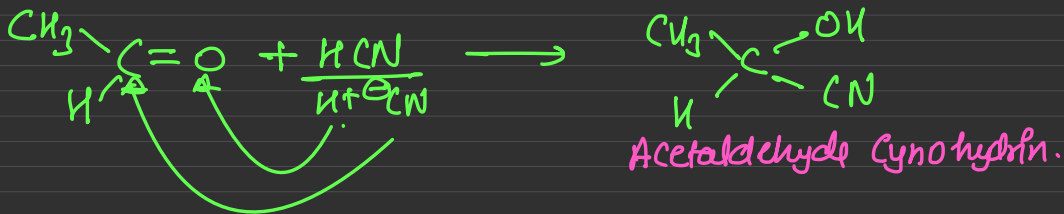
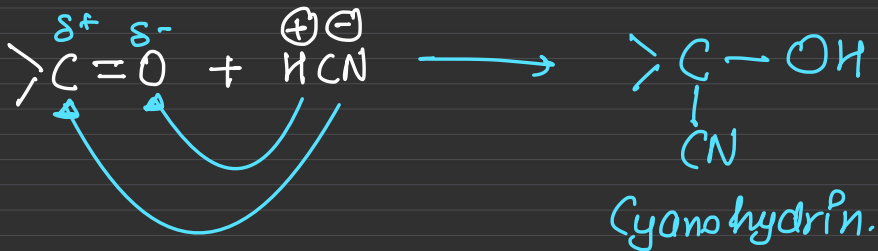
larger alkyl group in ketones
cause more steric hindrance so nucleophile can't
approach easily the carbonyl carbon

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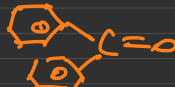
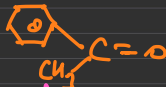
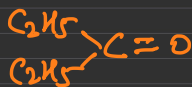
$\text{Nu}^\ominus$

Addition of HCN

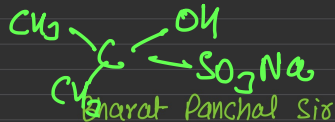
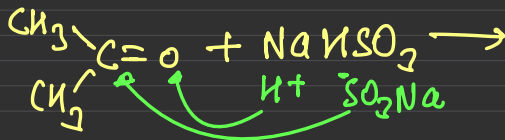
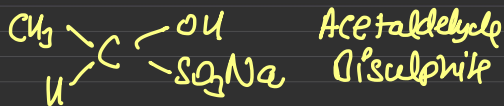
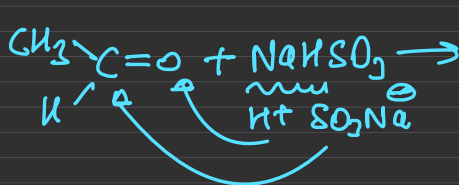


Addition of Sodium Bisulphite (NaHSO_3)

When aldehyde and ketone is treated with sodium bisulphite then fine white crystals are formed so this reaction is used as a test for carbonyl group.



Diethyl ketone, acetophenone, benzophenone do not give this reaction due to more steric hindrance



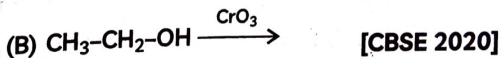
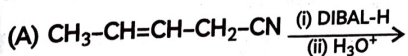
Sharat Panchal Sir

1. What happens when Benzene is treated with acetyl chloride in the presence of anhydrous AlCl_3 ? [CBSE 2020]

2. What happens when benzene is treated with CH_3COCl in presence of anhydrous AlCl_3 ? [CBSE 2020]

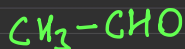
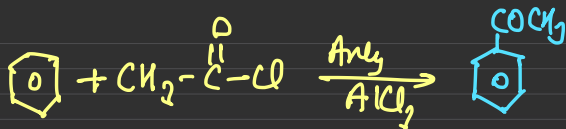
3. How can you convert the following : Benzene to acetophenone? [CBSE 2020]

Write the major product(s) in the following:

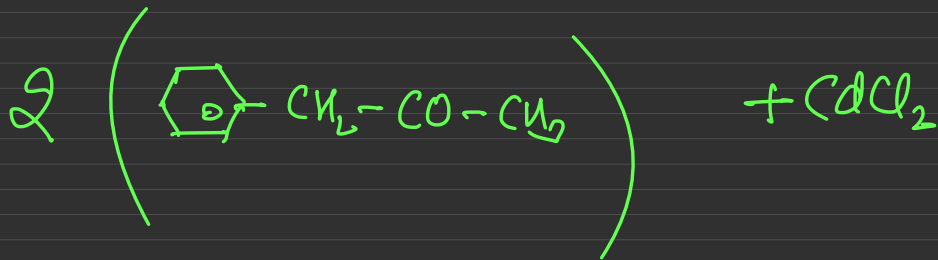


4. Write chemical equations for the following reaction: [CBSE 2019, 18]

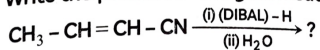
5. Complete the following reaction:



Complete the following reaction:



Write the product in the given reaction;



[CBSE 2017]

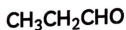
Write the structure of 2-methylbutanal.

[CBSE 2015]

Draw the structure of 3-methylpentanal.

[CBSE 2015]

Write the IUPAC name of the following:



[CBSE 2015]

How do you convert the following:

Ethyne to ethanal

[CBSE 2015]

Draw the structures of the following :

(A) *p* - Methylbenzaldehyde

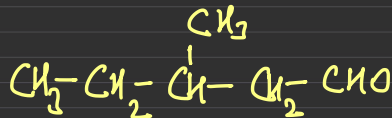
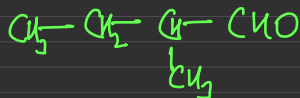
(B) 4 - Methylpent- 3-en-2-one [CBSE 2015]

Write the equations involved in the following reactions:

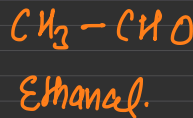
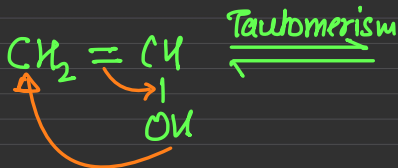
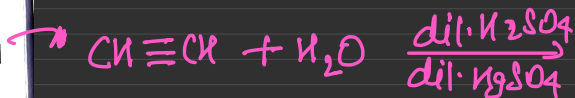
(A) Stephen reaction

(B) Etard reaction

[CBSE 2015]



Propanal



Draw the structures of the following :

(A) *p* - Methylbenzaldehyde

(B) 4 - Methylpent- 3-en-2-one [CBSE 2015]

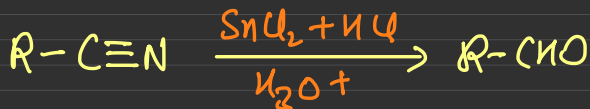
Write the equations involved in the following reactions:

(A) Stephen reaction

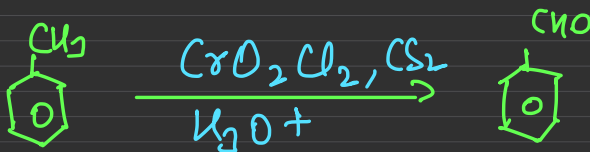
(B) Etard reaction

[CBSE 2015]

Stephen Rxn

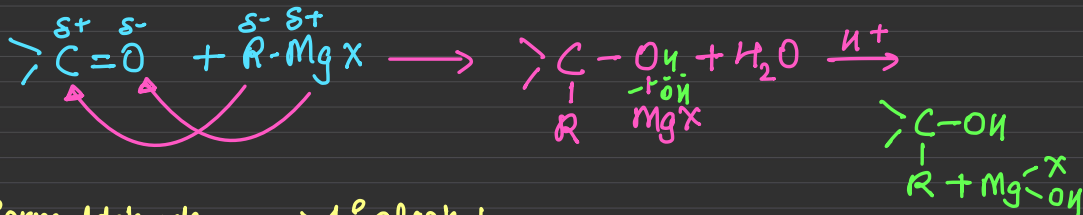


Etard Rxn



Bharat Panchal Sir

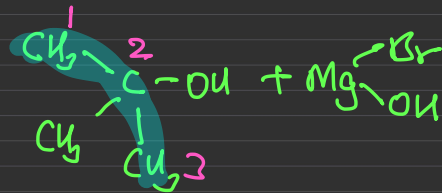
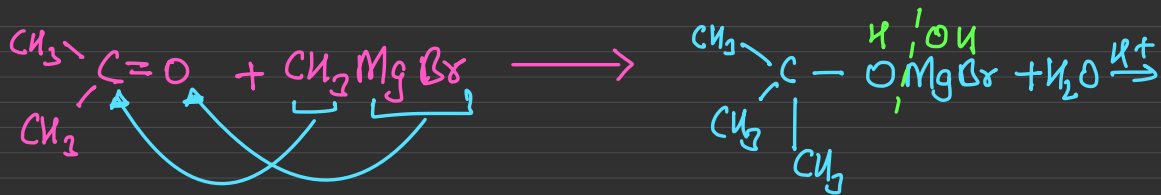
Rxn with Grignard Reagent



Formaldehyde $\longrightarrow$ 1° alcohol

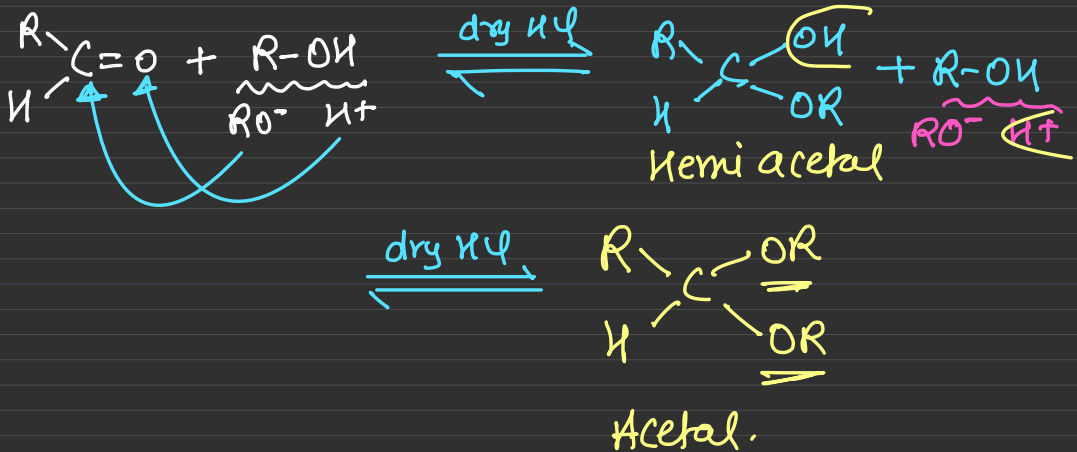
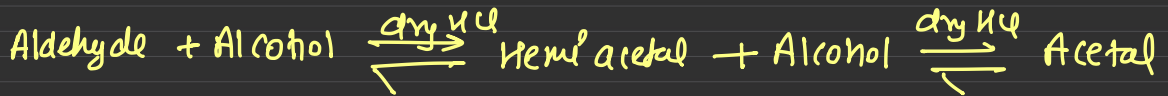
Aldehyde $\longrightarrow$ 2° alcohol

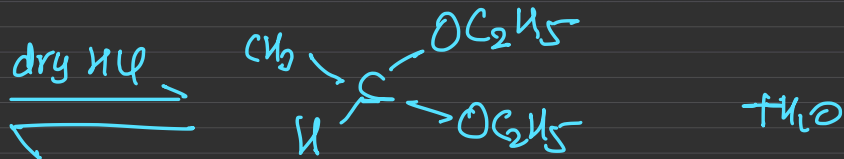
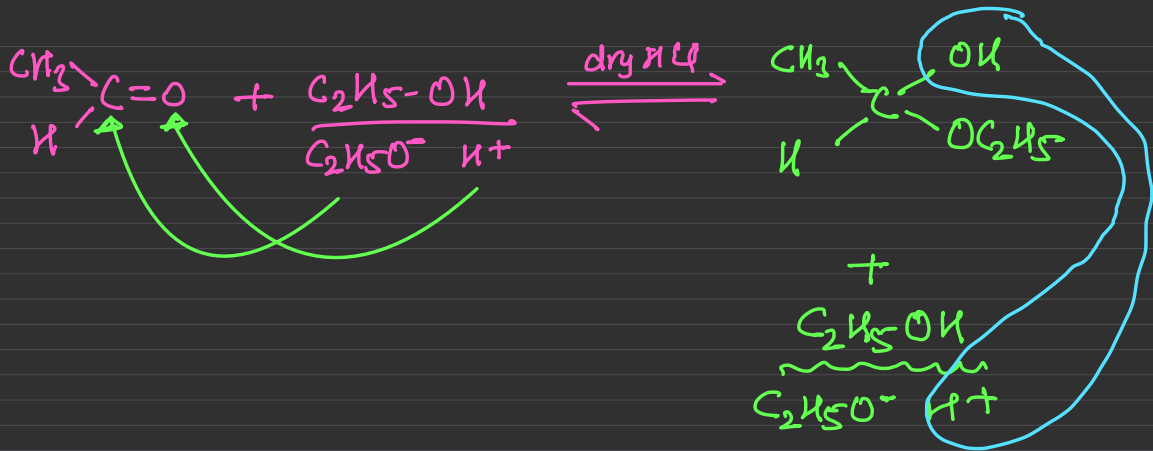
Ketone $\longrightarrow$ 3° alcohol

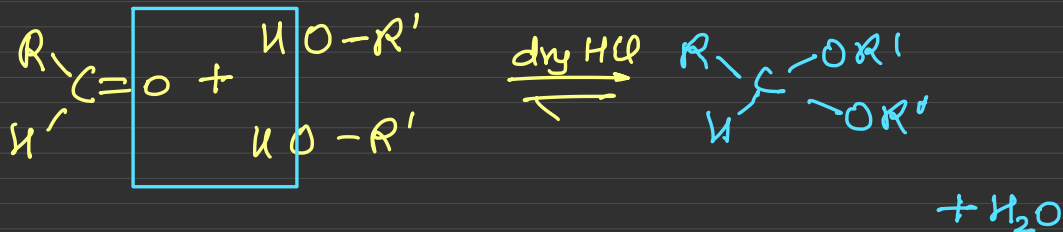
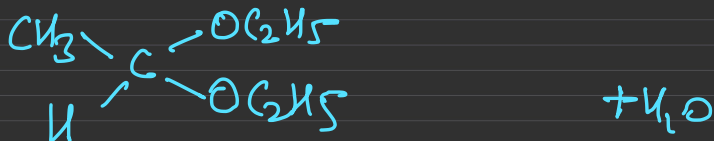
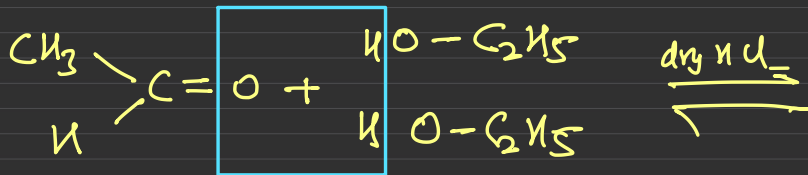


2-Methyl propan-2-ol
(3°)

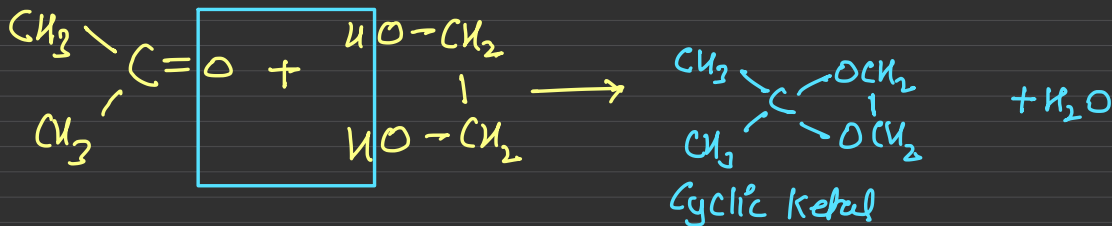
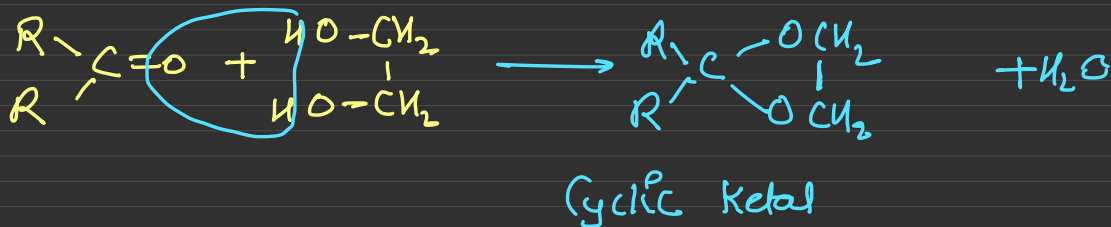
Addition of Alcohol







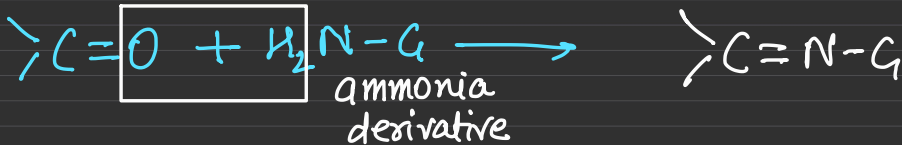
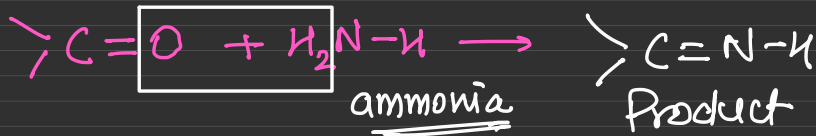
Note Ketones do not form acetal with monohydric alcohol, Although they form ketal with dihydric alcohol.



Nucleophilic Addition Rxn followed by loss of Water

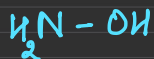
[NUCLEOPHILIC ADDITION ELIMINATION REACTION]

→ Ammonia and derivatives



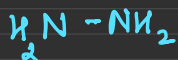
Ammonia derivative

Product



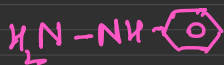
Hydroxyl amine

Oxime



Hydrazine

Hydrazone



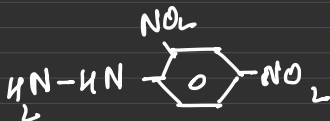
Phenyl hydrazine

Phenyl
hydrazone



Semi carbazide

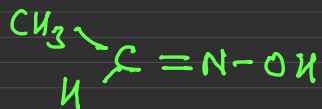
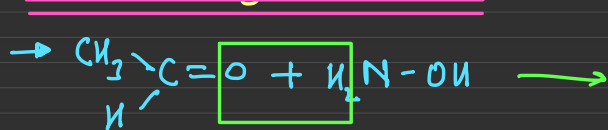
Semi Carbazone



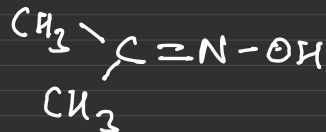
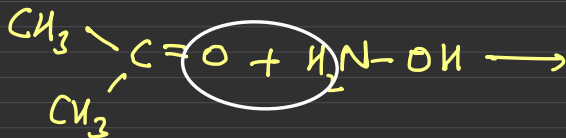
2,4 -Dinitro
phenyl hydrazine

2,4 -Dinitro
phenyl hydrazone

Rxn with hydroxylamine:



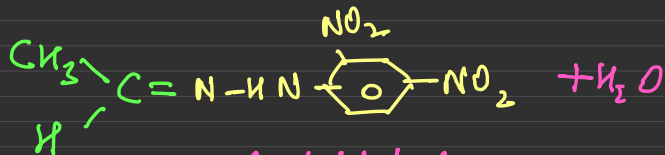
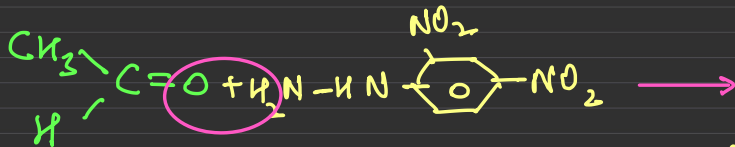
Acetaldehyde Oxime



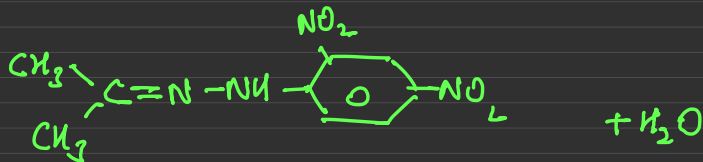
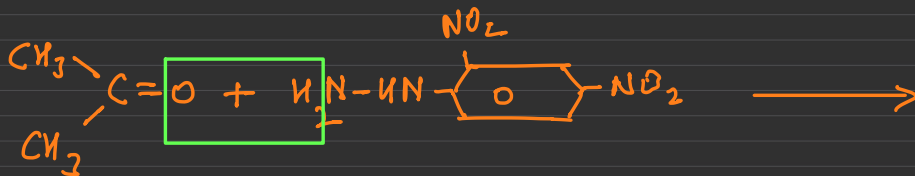
Acetone
Oxime

2,4-DNP Test

↳ For Carbonyl Groups → Orange coloured crystalline solid

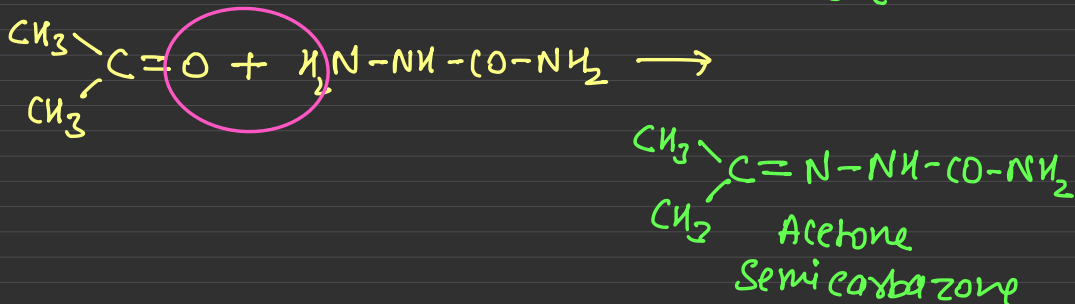
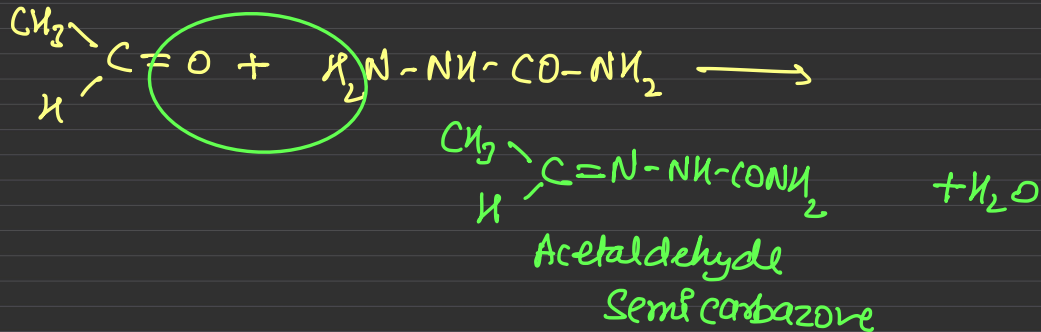


Acetaldehyde
2,4-Dinitrophenyl hydrazone
(Orange colour)



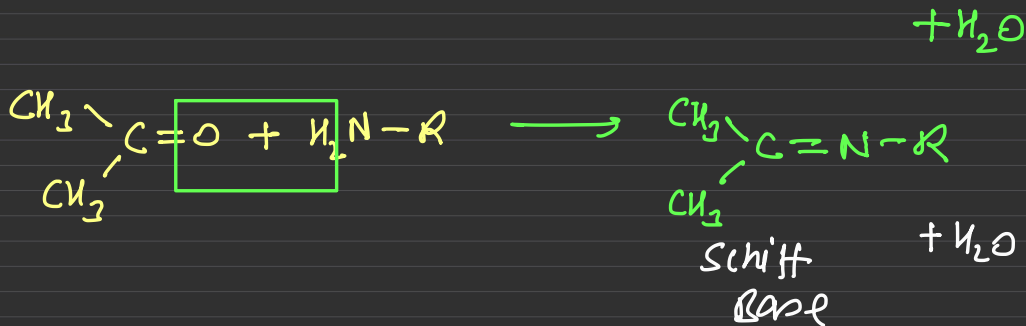
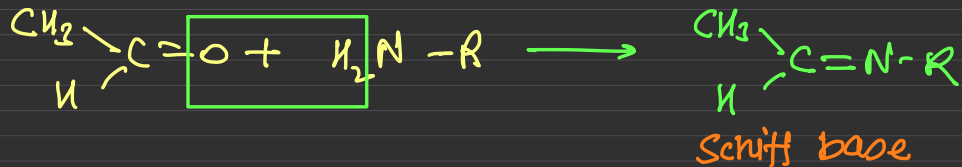
Acetone
 2,4-dinitrophenyl hydrazone

* Rxn with Semi Carbazide ($\text{H}_2\text{N}-\text{NH}-\text{CO}-\text{NH}_2$)



Bharat Panchal Sir

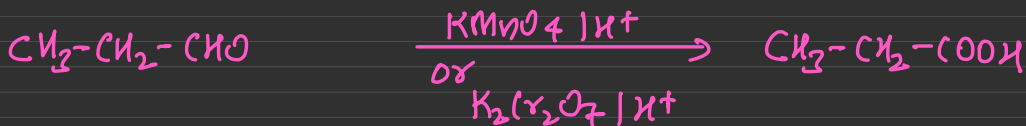
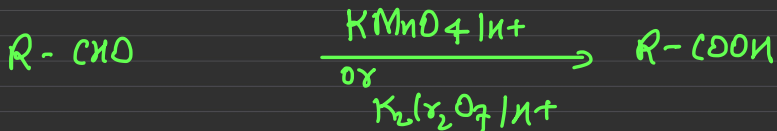
Rxn with Primary Amine



Oxidation

Aldehydes are easily oxidised to corresponding carboxylic acid.

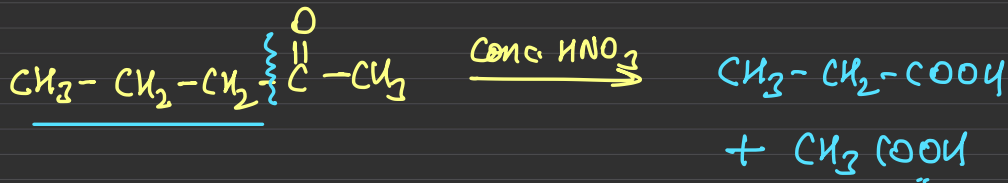
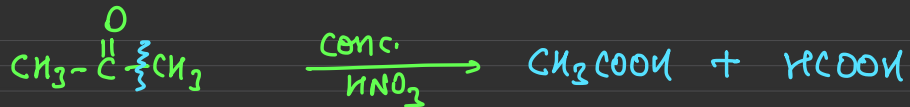
The oxidising agent used are acidic KMnO_4 or acidic $\text{K}_2\text{Cr}_2\text{O}_7$



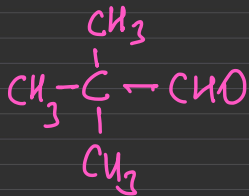
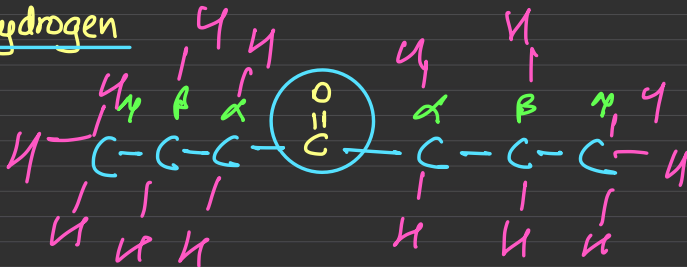
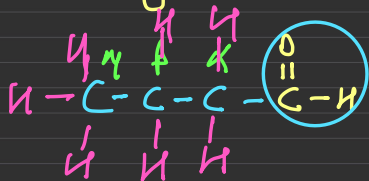
Note Ketones are not easily oxidised. But under drastic condition with powerful oxidising agent like conc. HNO_3 or KMnO_4 /conc. H_2SO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$ /conc. H_2SO_4

Poppoff's Rule

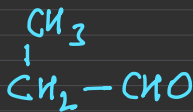
In an unsymmetrical ketone, during oxidation, the breaking of C-C bond occurs in such a way that keto group goes with smaller alkyl group.



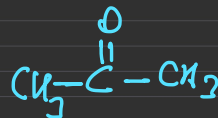
Reactivity due to α -hydrogen



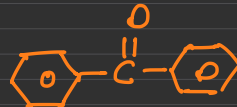
no α -hydrogen



have 2 α -H

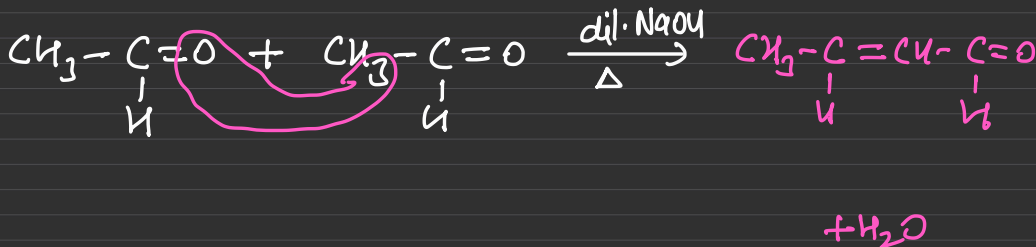
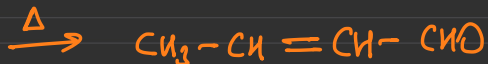
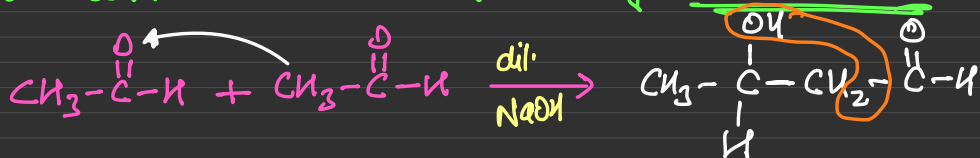


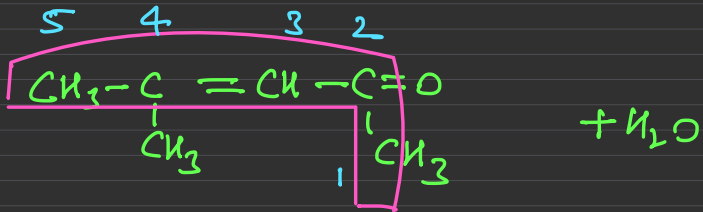
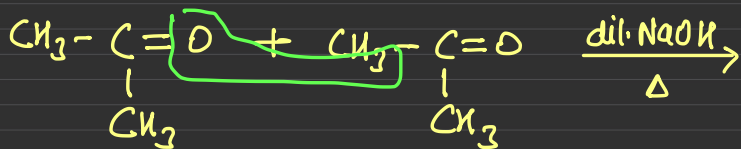
No α -H



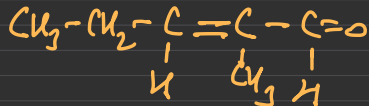
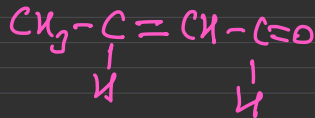
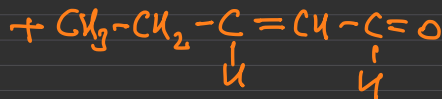
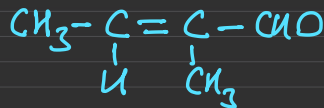
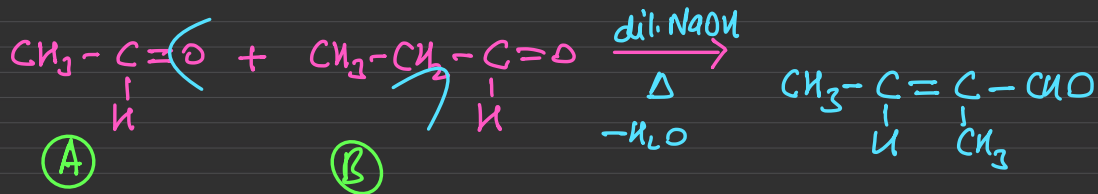
ALDOL CONDENSATION Rxn

Those aldehyde and ketones have α -hydrogen undergo aldol condensation in the presence of dilute alkali





4-Methylpent-3-en-2-one



4 products

A + B

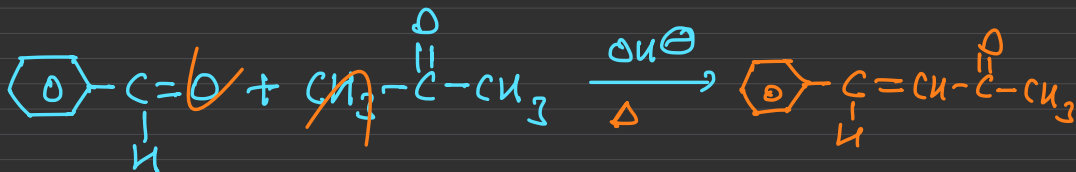
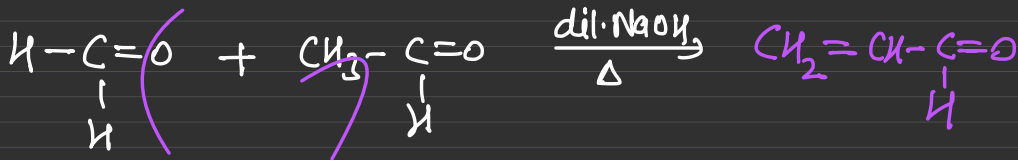
B + A

A + A

B + B

Cross Aldol Condensation

- different carbonyl group (one of them must have α -hydrogen)
- dil NaOH or dil Ba(OH)₂

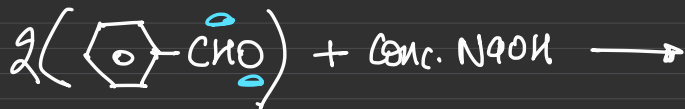




Bharat Panchal Sir

CANNIZZARO'S REACTION

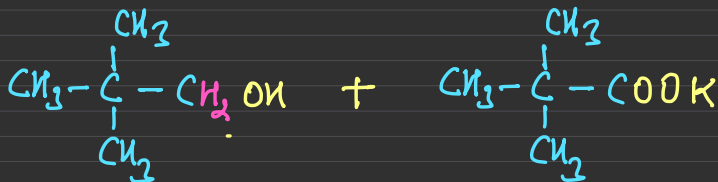
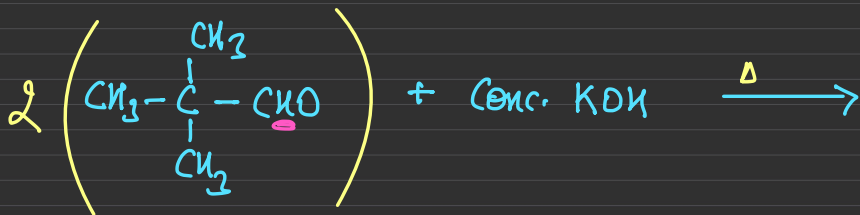
- ↳ Those aldehyde which do not have α -hydrogen
- ↳ Conc. alkali [conc. NaOH or conc. KOH]
- ↳ Disproportionation reaction [Self oxidation + Self Reduction]



Redn



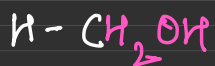
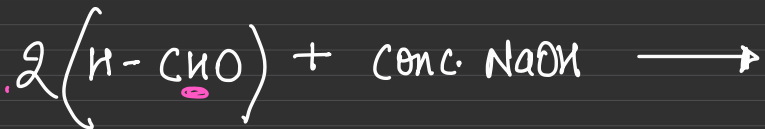
Oxi



Redn

Oxidn

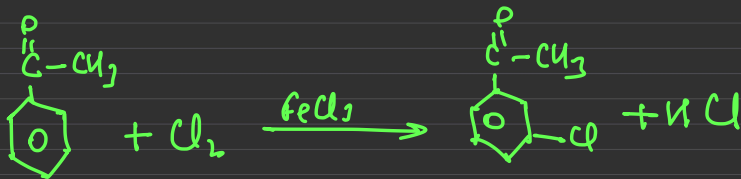
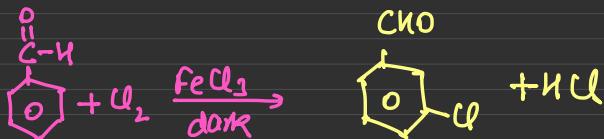
Bharat Panchal Sir



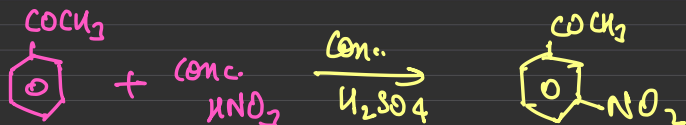
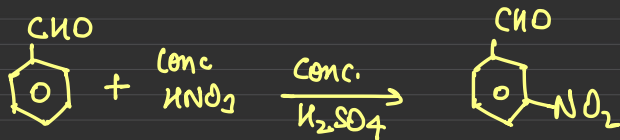
Electrophilic Sub. Rxn

$\text{C}(=\text{O})-\text{H}$, $-\text{C}(=\text{O})-$ deactivating group
These are meta directing.
doesn't undergo Friedel-Craft Alkylation and Acylation

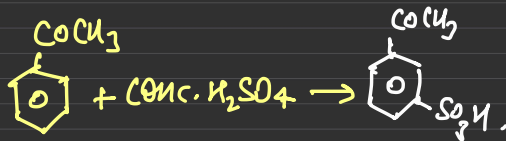
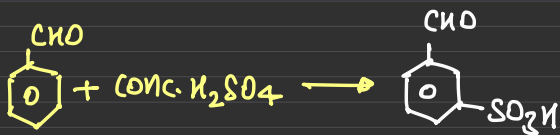
Chlorination

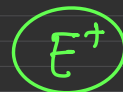
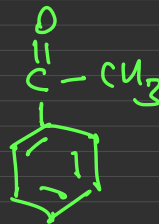
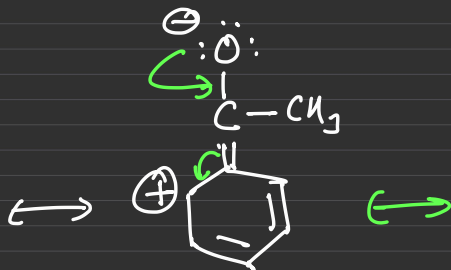
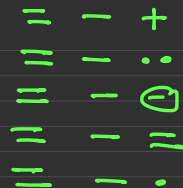
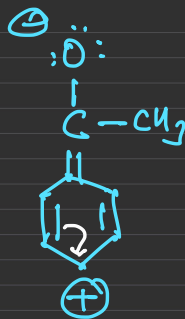
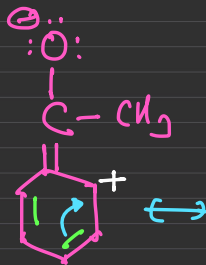
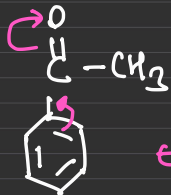


Nitration



Sulphonation





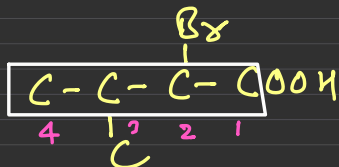
Bharat Panchal Sir

CARBOXYLIC ACID

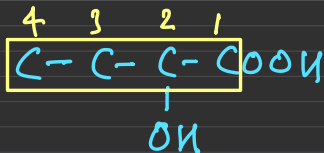
-COOH

-oic acid

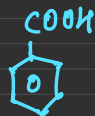
H-COOH methanoic acid



2-bromo-3-methyl
butanoic acid.

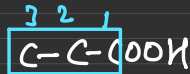


2-hydroxybutanoic acid

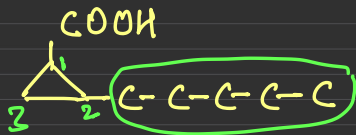


Benzene carboxylic
acid

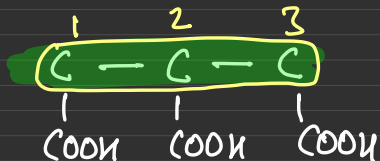
or Benzoic acid



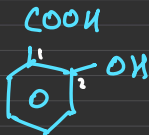
3-Cyclopentyl
propanoic acid



2-pentyl cyclopropane carboxylic acid



Propane-1,2,3-tricarboxylic acid

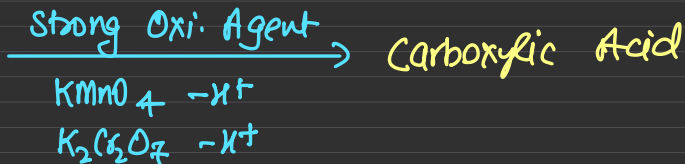


2-hydroxy benzoic acid

Preparation

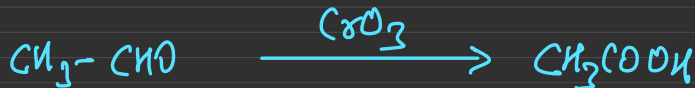
↳ from Alcohol

1° alcohol



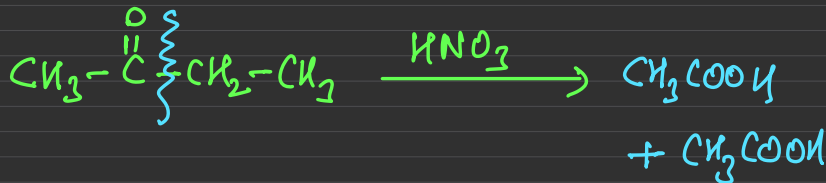
From Aldehyde and Ketone

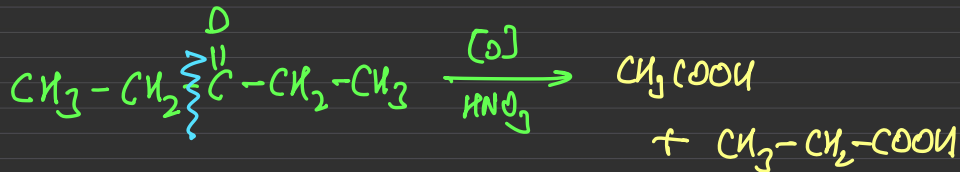
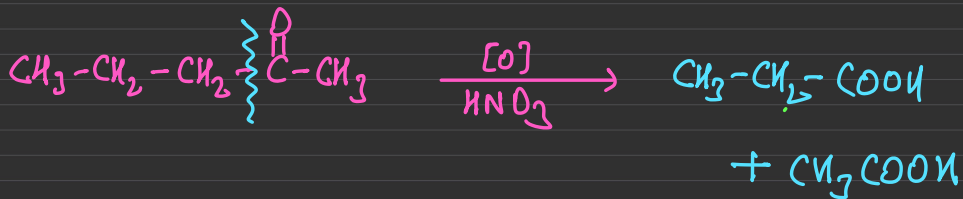
Aldehyde $\xrightarrow{\text{Mild Oxi. Agent}}$ Carboxylic acid



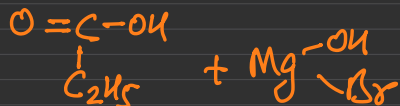
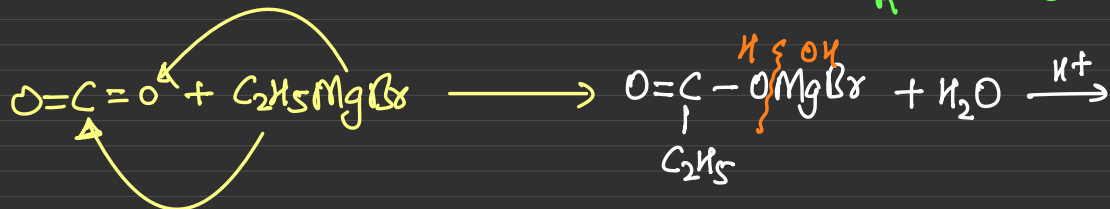
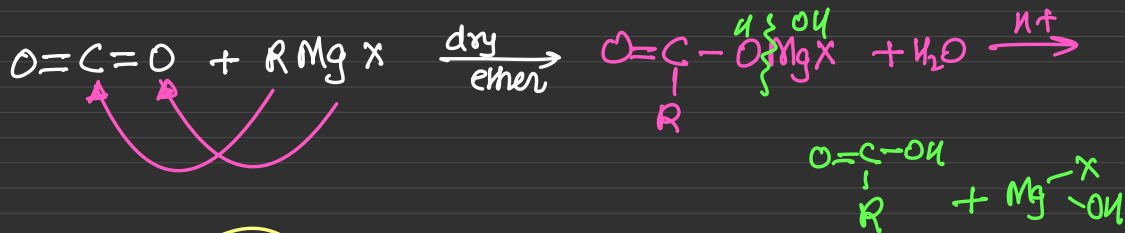
Ketone $\xrightarrow[\text{HNO}_3]{\text{strong Oxi. Agent}}$ Carboxylic Acid

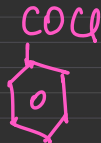
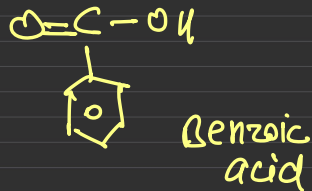
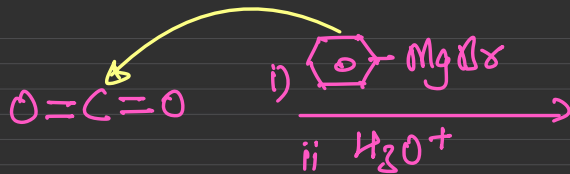
→ Pott Rule follow



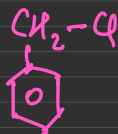


from Grignard Reagent



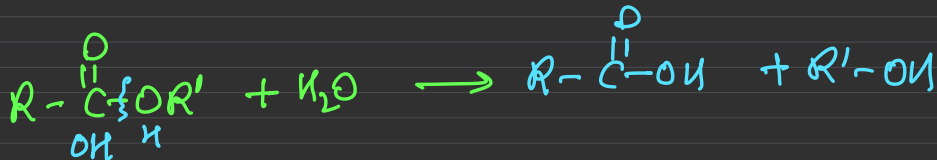
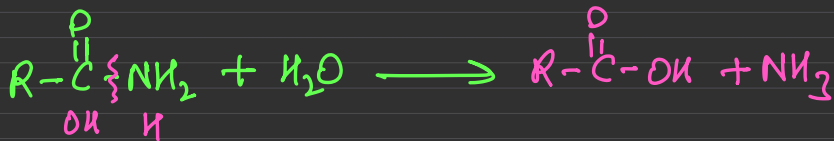
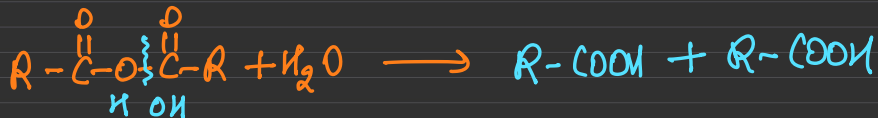
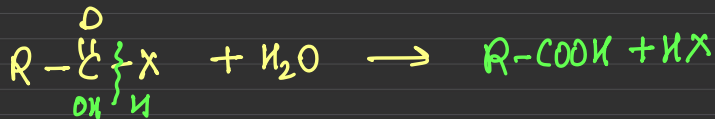


Benzoyl
chloride



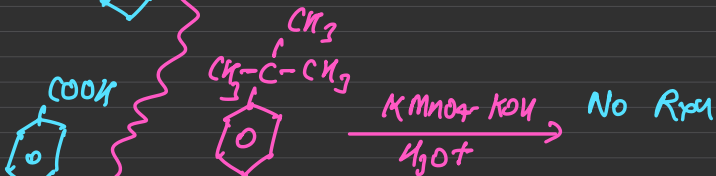
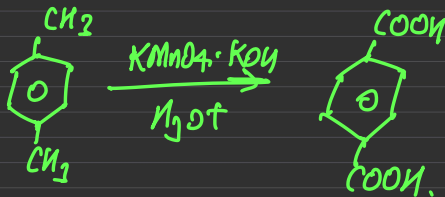
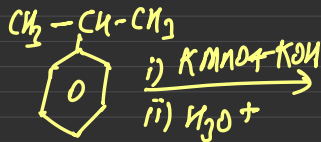
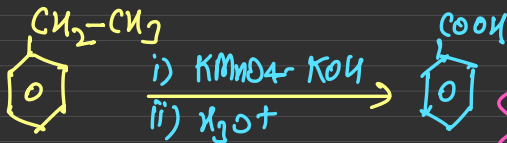
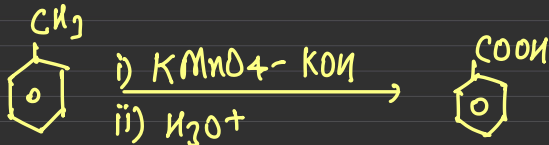
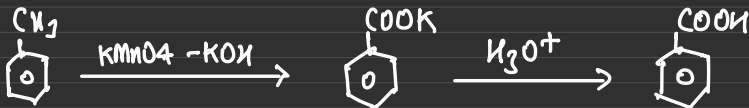
Benzyl
chloride

from Acid Halide, Acid Anhydride, Acid Amide and Ester :



99-997.

from Alkyl Benzene

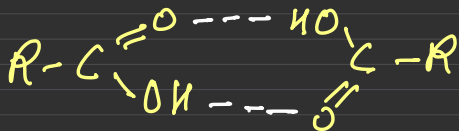


As no benzylic hydrogen is present
Bharat Panchal Sir
So no oxidation takes place

PHYSICAL PROPERTIES

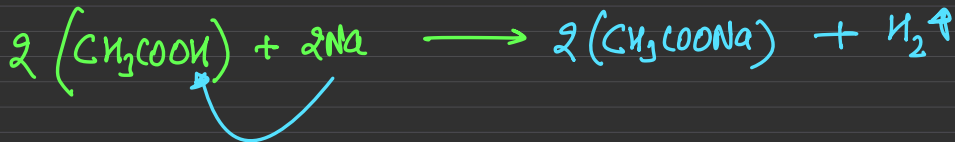
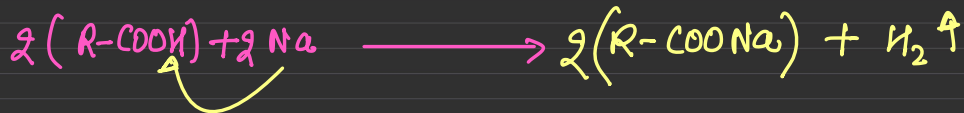
→ Solubility First four members of aliphatic carboxylic acid are soluble in water due to the formation of H-bond. As mol. mass increases solubility in water decreases due to hydrophobic part.

→ Boiling Point Due to the presence of hydrogen bond, boiling point of carboxylic acids are higher than corresponding aldehyde, ketone, and even alcohol.

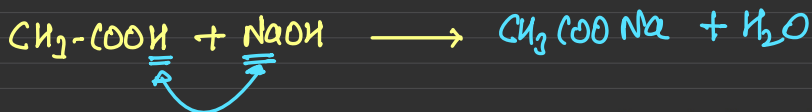
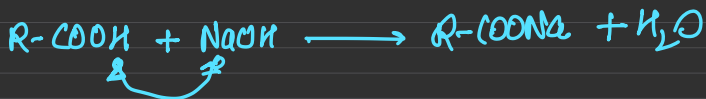


CHEMICAL PROPERTIES

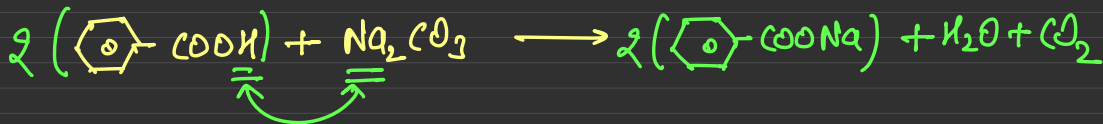
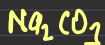
• Rxn with metal



→ Rxn with Alkali

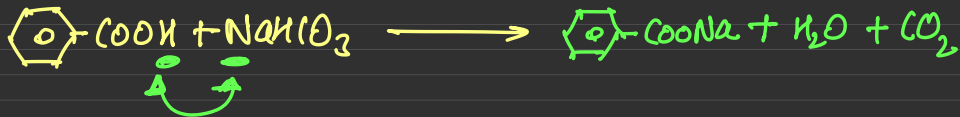
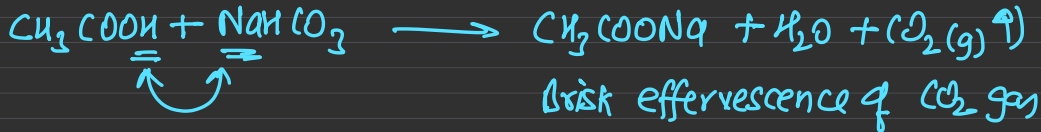


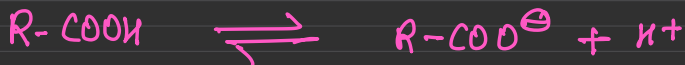
R^u with carbonate and bicarbonate





NaHCO₃ Test





$$K_a = \frac{[R-COO^{\ominus}][H^{+}]}{[R-COOH]}$$

$K_a \rightarrow$ dissociation
constant of Acid

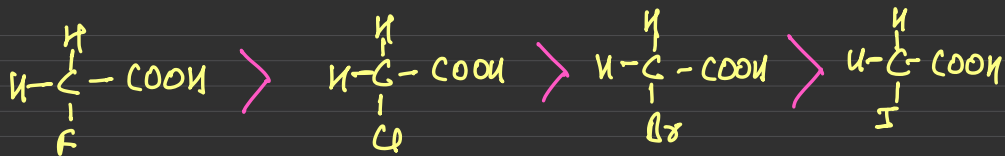
$$pK_a = -\log K_a$$

$K_a \rightarrow$ acidic strength

$pK_a \rightarrow$ $\frac{1}{\text{acidic strength}}$

E.W.G increase acid strength

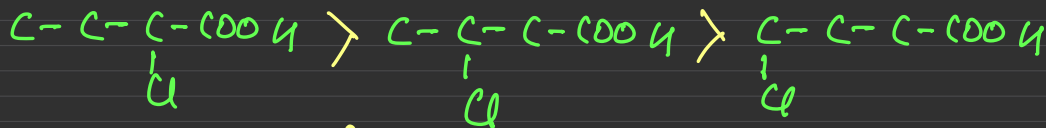
E.D.G decrease acid strength



as Electronegativity increase, acidic strength increase



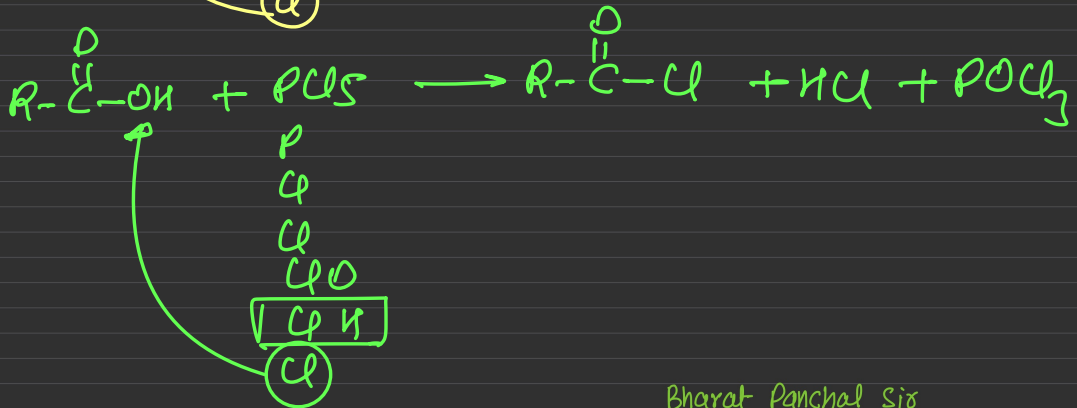
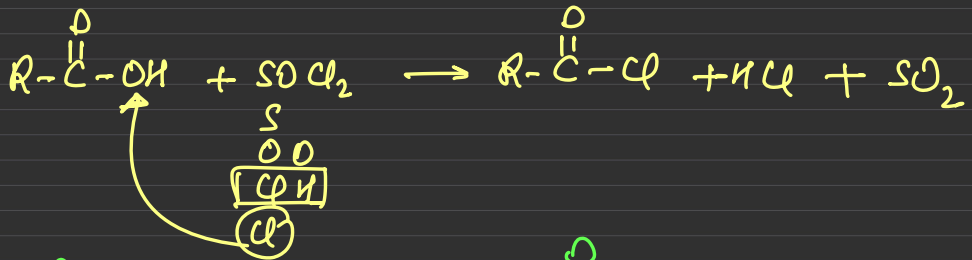
as no. of E.W.G increases, acidic strength increases



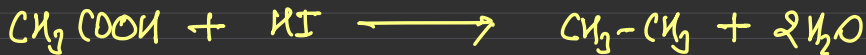
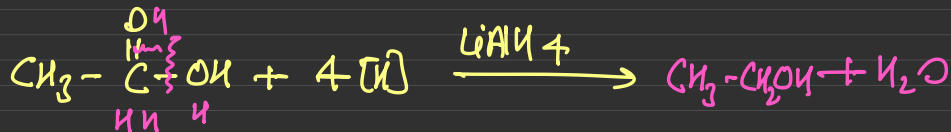
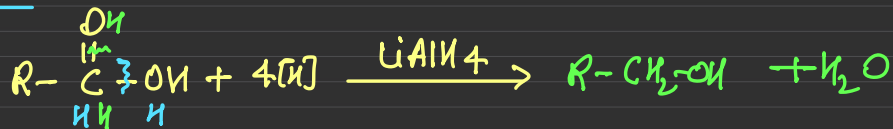
as distance of E.W.G increase
acidic strength decreases

Bharat Panchal Sir

Rxn with PCl_5 , PCl_3 , SOCl_2

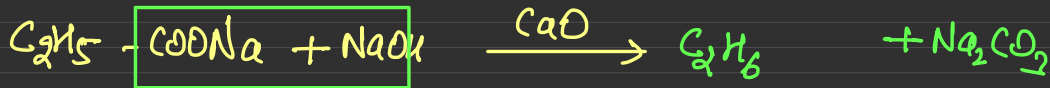


Reduction

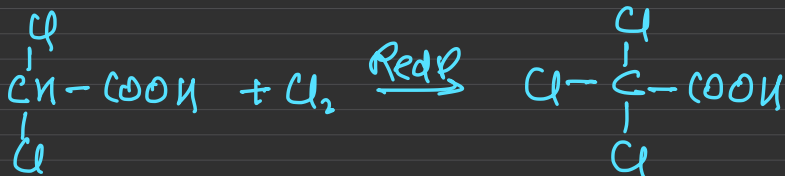
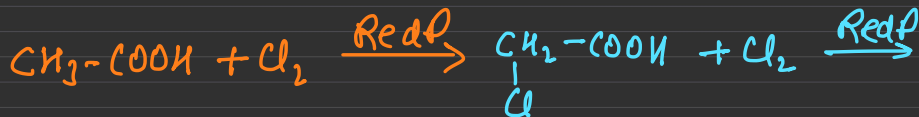


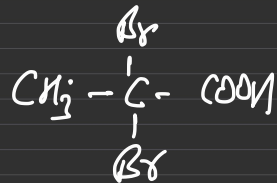
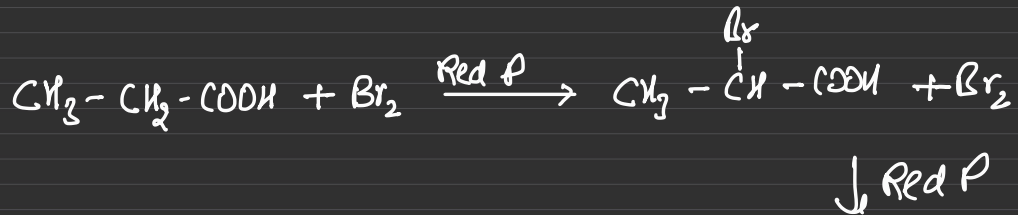
DECARBOXYLATION Rⁿn

Soda $\rightarrow$ NaOH
lime $\rightarrow$ CaO

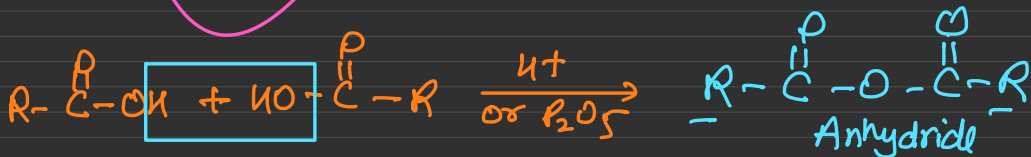
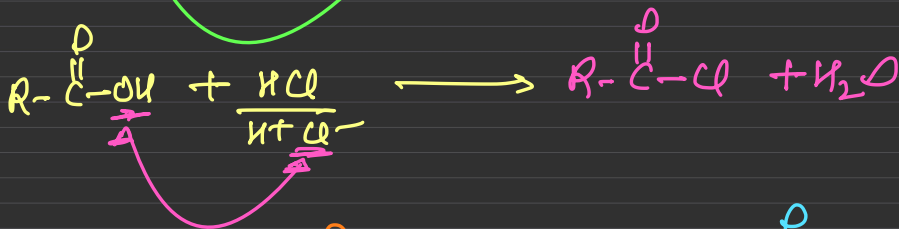
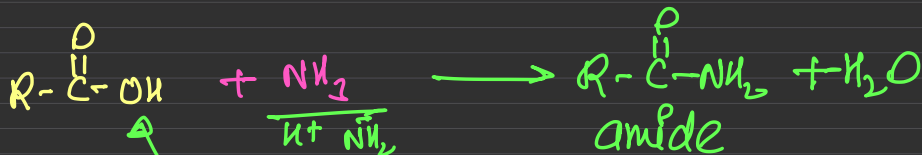
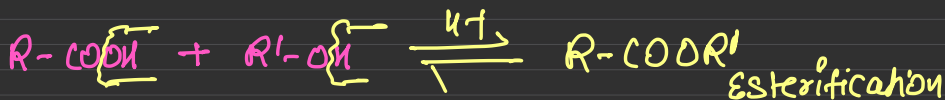


$\Rightarrow$ H.V.Z Rⁿn (Hell Volhard Zelinshy Rⁿn)





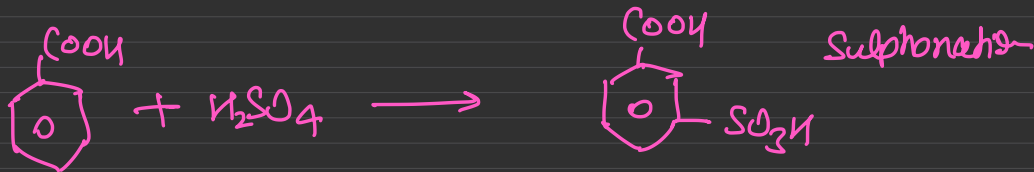
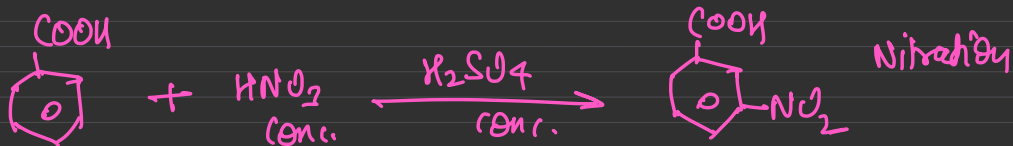
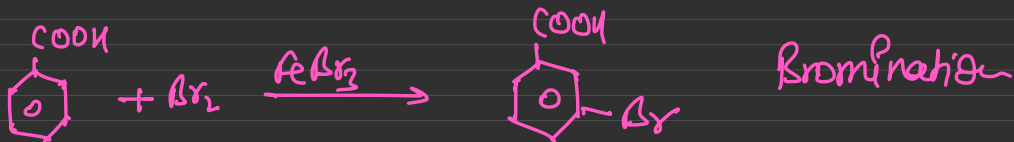
Carboxylic Acid Derivative formation



Bharat Panchal Sir

Electrophilic Sub. Rxn

- $\overset{\text{O}}{\parallel}\text{C}-\text{OH}$ group is a deactivating group,
meta directing



Thank
You



Bharat Panchal Sir